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FINAL Report

EE3001 TELECOMMUNICATION SYSTEM DESIGN USING DSP

CAPSUL PROJECT

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1. ABSTRACT

In this experiment, a comprehensive digital communications system is created in MATLAB in order to process, encode parameter-wise, convert to binary code, BPSK modulate, and transmit/receive a distorted speech signal, originally sampled at a rate of 48 kHz and affected by a high-pass filter, over a single USRP-2900 (B200) software-defined radio using an actual radio frequency loopback. In the processing stage of the signal, an FIR Hamming low-pass filter of a sufficiently high order together with the decimation operation are applied in order to reduce the sampling rate to 8 kHz. For speech analysis, an LPC representation of order 14, according to the model of source-filter speech production is used; in order to increase the stability of encoding and decrease the quantization errors, a transformation to LSF representation is introduced. The encoded representation consists of 128 bits per each frame that include LSF vector, log-gain, pitch and voice/non-voice markers. BPSK modulation with pulse-shaping by RRC function with roll-off coefficient of 0.35 is employed for transmission in the 915 MHz ISM band. Reception involves matched filtering, synchronization by complex cross-correlation, and hard-demodulation followed by speech reconstruction by means of the 1/A(z) filter. The system functions successfully with a measured bit error rate on the order of 10^{-4}, an error-free reconstruction of 51 out of 52 frames, and a 10.4:1 compression ratio relative to 16-bit PCM, producing intelligible reconstructed speech.

2. OBJECTIVE

In the current research, the issue of parametric speech representation at a low bit rate and subsequent reliable transmission over a real radio frequency channel was considered. This issue was addressed by combining the processes of speech source coding (compression), modulation, and digital transmission in a unified end-to-end approach. The work was conducted using the MATLAB DSP toolbox, the Communications Toolbox SDR interface, and the real USRP-2900 (B200) software-defined radio.

The traditional 16-bit pulse-code modulation speech encoding method requires a 128-kbps bitrate at an 8kHz sampling rate, which is rather high for efficient transmission over limited bandwidth communication channels. In comparison, the parametric linear prediction coding (LPC) method that employs the speech model based on the vocal tract (vocoder approach) uses just a few informative parameters to represent speech (spectral envelope, gain, pitch, voiced/unvoiced decision), and thus it provides a much lower bitrate for intelligible speech. Thus, the parametric encoding plays an important role in bandwidth-limited communication systems; namely, reduction in the number of transmitted bits enhances the spectrum efficiency and enables more simultaneous users to operate within the same bandwidth.

The aim of this project is to realize the previously described chain of operations, i.e., the parametric coding-modulation-transmission-reconstruction scheme, in practice, and study the impact of the channel and of the hardware in the communication link on system performance. The objectives of the project are as follows:

•To produce a noise-free speech signal with sampling rate 8 kHz from the noisy speech signal sampled at 48 kHz by means of anti-aliasing and decimation.

•To parametrize the speech in terms of LPC/LSF, gain, pitch, and V/UV components and quantize them into 128 bits per frame.

•To modulate the resulting binary data sequence using RRC-shaped BPSK, and transmit it over the USRP-2900 in the 915 MHz band of ISM.

•To perform synchronization, demodulation and LPC reconstruction of the speech on the receiver side, and analyze the performance of the system in terms of BER, SNR, EVM, and compression ratio.

3. BACKGROUND

3.1 Source-Filter Speech Production Model and LPC

Human speech production is classically represented by the source-filter model. In this model, an excitation source (a quasi-periodic pulse train produced by the vocal cords for voiced sounds, and white noise corresponding to turbulent airflow for unvoiced sounds) is passed through a linear, slowly time-varying filter representing the vocal tract. Linear Predictive Coding (LPC) is based on the principle of modeling this vocal-tract filter with an all-pole transfer function. In the LPC model, the current speech sample is predicted as a linear combination of past samples:

$$s[n] \approx \Sigma\, a_k \cdot s[n-k]\,, \quad k = 1 \ldots p \tag{1}$$

Here, the coefficients a_k are the LPC prediction coefficients and p is the model order (in this study, the standard value of p = 14 was chosen for 8 kHz). These coefficients are estimated using the autocorrelation method (the Levinson-Durbin recursion), which minimizes the energy of the prediction error. The resulting synthesis filter is all-pole in the form 1/A(z) and models the resonance (formant) structure of the vocal tract.

3.2 Rationale for the LSF (Line Spectral Frequencies) Transformation

However, the LPC coefficients (a_k) are not well-suited for quantization due to the possibility that relatively minor quantization inaccuracies will make the synthesis filter unstable (by pushing poles out of the unit circle) and also due to the large dynamic range of the LPC coefficients reducing quantization performance. The LPC coefficients were therefore transformed into LSF (Line Spectral Frequencies), which offer superior quantization performance. The LSF parameters consist of the zeros of the symmetric part (P(z)) and antisymmetric part (Q(z)) of the A(z)

polynomial, with all parameters located on the unit circle in an interlaced (ordered) fashion. As such, the LSF representation offers two theoretical advantages when it comes to quantization. Firstly, provided that the ordering of quantized LSF coefficients is maintained, the synthesis filter will remain stable. Secondly, since the connection between spectral error and LSF parameters is localized, quantization inaccuracies can be kept to a minimum.

3.3 BPSK Modulation, RRC Pulse Shaping, and Matched Filtering

BPSK (Binary Phase-Shift Keying) is the form of 1-Dimensional Modulation, where each symbol is represented by either of the two antipodal phases of the carrier (0° or 180°), which in effect represent the amplitudes ±1. With the 1-dimensional decision making space, BPSK stands out to be one of the most robust forms of modulation providing minimum error rates given a specific Eb/N0. While transmitting, a Pulse Shaping scheme is implemented in order to limit Inter-Symbol Interference and to control the occupied bandwidth. In this research, Root-Raised Cosine Filter meeting the requirements of Nyquist ISI Criterion (with equal division of power between transmitter and receiver parts) having roll-off of 0.35 was used. At the receiver, another filter with same characteristic but Root Raised Cosine shape (equal in size to the transmitter filter) was applied to implement a matched filter. The multiplication of the two RRC filters gives a complete Raised Cosine filter minimizing ISI at the sampling points and giving maximum SNR at the receiver.

In case of AWGN, the expression for BPSK Bit Error Rate (BER) vs Eb/N0 is provided below:

$$BER_theory = \tfrac{1}{2} \cdot erfc(\sqrt{(10\text{^}(SNR_dB / 10))}) \quad (2)$$

This expression was used as the reference for comparing the measured performance against the theoretical expectation.

4. DESIGN AND TEST PROCEDURES

The system was designed as a methodological chain consisting of the stages of source coding, modulation, hardware transmission, demodulation, and synthesis. The chain begins with the DSP front-end at the transmitter, passes through the USRP hardware layer, and is completed by the DSP back-end at the receiver. The overall processing flow is as follows:

Corrupted Speech (48 kHz) → Signal Preparation (LPF + Decimation, 8 kHz) → LPC/LSF Analysis → Quantization (128 bits/frame) → BPSK + RRC + Preamble → USRP TX (915 MHz) → USRP RX → Matched Filter → Correlation Synchronization + Phase Correction → BPSK Hard Decision → Inverse Quantization → LPC Synthesis (OLA) → Reconstructed Speech (8 kHz)

The fundamental system parameters used in the design are summarized in Table 1. Each sub-process is described step by step with technical details in the following subsections.

Table 1. Fundamental system parameters.

Parameter	Value	Description
Input sampling rate	48 kHz	Corrupted input signal
Speech processing rate	8 kHz	Decimation factor = 6
Frame length	160 samples	20 ms
Frame shift (hop)	80 samples	10 ms, 50% overlap
LPC order	14	Standard @ 8 kHz
Pre-emphasis coefficient	0.97	High-frequency emphasis
Pitch search range	20–147 samples	54–400 Hz
Symbol rate	50 kBd	BPSK: 1 bit/symbol
Samples/symbol	4	Oversampling factor
RRC roll-off	0.35	Pulse shaping
Carrier frequency	915 MHz	ISM band
USRP sampling rate	200 kHz	50 kBd × 4
Master clock rate	20 MHz	Interp./Decim. = 100
TX / RX gain	70 / 50 dB	Wireless mode
Preamble length	64 bits	Pseudo-random (rng=338)

4.1 Signal Preparation

A corrupted speech signal sampled at 48 kHz and containing unwanted high-frequency components was provided as the system input. Since the LPC/LSF analysis stage is defined at an 8 kHz sampling rate, the input signal must be decimated by a factor of 6. The decimation factor was determined as follows:

$$Decimation\ factor = f_input / f_speech = 48000 / 8000 = 6 \quad (3)$$

To prevent aliasing distortion, a 180th-order, Hamming-windowed, linear-phase FIR low-pass filter (LPF) was designed prior to decimation, with its cutoff frequency selected below the new Nyquist limit (≈4 kHz). To eliminate phase distortion, the filter was applied in a zero-phase manner (filtfilt). After filtering, the signal was decimated by a factor of 6 to an 8 kHz sampling rate, and finally its amplitude was normalized to the [−1, 1] range. This step is a prerequisite for the numerical stability of the subsequent LPC analysis.

4.2 LPC/LSF Analysis and Parameter Extraction

A pre-emphasis filter that boosts high frequencies was first applied to the 8 kHz speech signal. This filter has the transfer function $(1 - 0.97 \cdot z^{-1})$ and enables the LPC analysis to model the spectral envelope more smoothly. The signal was divided into windows of 20 ms (160 samples),

and the windows were processed with 50% overlap using a 10 ms (80-sample) shift. For each frame, 14th-order LPC analysis was performed to estimate the spectral envelope, and the resulting LPC coefficients were converted to the LSF representation; invalid (unordered or out-of-range) LSF sets were replaced with a default stable set. In addition, the pitch period was estimated from the peak location of the normalized autocorrelation within the 20–147 sample range, and the frame was classified as voiced or unvoiced (the V/UV decision) by comparing the peak value against a threshold of 0.35. The gain was obtained from the RMS value of the pre-emphasized frame in the logarithmic (dB) domain.

4.3 Quantization and Frame Structure

The parameters extracted for each frame were quantized within a fixed 128-bit budget. The total number of bits per frame is computed as follows:

$$bits/frame = 14\times 8\ (LSF) + 8\ (Gain) + 7\ (Pitch) + 1\ (V/UV) = 128\ bits \tag{4}$$

The allocation of the bit budget across the fields is given in Table 2. The LSF coefficients were passed through a sorting operation that guarantees stability prior to quantization, while the gain was quantized in the perceptually more appropriate logarithmic (dB) domain.

Table 2. Allocation of the 128-bit per-frame budget across fields.

Field	Quantization method	Bit count
LSF (14 coeffs)	Each LSF mapped from (0, π) to 8 bits	112
Log-Gain	Log-domain, clipped to [−60, 0] dB	8
Pitch	20–147 sample range to 128 levels	7
V/UV	Voiced/unvoiced binary flag	1
Total	**Per frame**	**128**

4.4 BPSK Modulation and the USRP Hardware Layer

A 64-bit pseudo-random synchronization preamble was prepended to the quantized bit stream to enable frame synchronization at the receiver. The resulting binary sequence was converted to ±1 symbols via the $\{0,1\} \rightarrow \{-1,+1\}$ BPSK mapping, then upsampled with 4 samples per symbol and shaped with an RRC filter having a roll-off of 0.35 and a span of 8 symbols. The signal was normalized to a peak value of 1 to preserve the USRP DAC range. The symbol rate was realized as 50 kBd and the occupied bandwidth as 67.5 kHz.

The shaped baseband signal was streamed to the USRP-2900 (B200, serial: 31AE1A6) hardware. The hardware was operated in the 915 MHz ISM band at a 200 kHz USRP sampling rate with a 20 MHz master clock; the transmitter carried the baseband signal by 100× interpolation, while the receiver collected it back from the RX2 antenna with 100× decimation. The RF up conversion and down conversion are done within the USRP board itself. Thus, the signals obtained in MATLAB are also represented at baseband frequencies of 0 Hz, which physically represent the carrier frequency of 915 MHz.Therefore, the bandwidth of around ±34 kHz gets shifted in the range of 914.97 to 915.03 MHz. The transmission was carried out via an interleaved TX/RX loopback mechanism on a single radio: the TX signal was split into 4096-sample chunks, an RX call was issued after each TX chunk, and once the transmit stream ended, a 20-chunk drain operation was applied to empty the receiver buffer, ensuring that the final portion of the transmission was also captured in full. The hardware parameters are summarized in Table 3.

Table 3. USRP-2900 hardware parameters.

Hardware parameter	Value
Platform / Serial no.	B200 / 31AE1A6
Master clock rate	20 MHz
Interpolation / Decimation	100 / 100
USRP sampling rate	200 kHz
Center frequency	915 MHz (ISM)
RX frame size	4096 samples
Receive antenna port	RX2
TX gain / RX gain	70 dB / 50 dB

4.5 Receiver Chain and Demodulation

At the receiver, the received signal was first passed through the same RRC filter as a matched filter and downsampled to 1 sample/symbol. The DC offset was removed and the signal normalized. To detect the frame start, the complex cross-correlation between the known 64-bit preamble and the received signal was computed; the correlation peak yielded the frame alignment. The ratio of the correlation peak value to the noise floor is computed as a metric that quantifies the synchronization quality:

$$corr\ SNR = 20 \cdot \log_{10}(peak / noise_floor) = 31.5\ dB \quad (5)$$

The 180° phase ambiguity inherent in BPSK transmission was resolved by rotating the signal according to the phase of the correlation output (phase correction); the real part of the signal was used for the decision, while the complex form was preserved for the constellation plot. The aligned symbols were sampled at the symbol rate and converted back to binary form by a hard decision

against a zero threshold, and inverse quantization was applied to recover the LSF, gain, pitch, and V/UV parameters of each frame.

4.6 LPC Speech Synthesis

The speech was synthesized based on the recovered parameters on a per-frame basis. Prior to synthesizing the speech, the LSF parameter was converted to the LPC parameter, and the stability was checked whether poles lie inside the unit circle before designing the 1/A(z) synthesis filter. Depending on the V/UV decision made, the excitation was synthesized in such a way that if a frame is a voiced signal, then an impulse train will be used as the excitation based on the pitch value, whereas if it is an unvoiced signal, then the excitation would be a random noise sequence having an energy balance with the voiced signal √(pitch). The resulting excitation was scaled with the gain and then filtered with the 1/A(z) synthesis filter; the overlapping windows are added using the OLA (overlap-add) method. Finally, the inverse of the analysis-stage pre-emphasis filter was applied, and the output was normalized to a peak value of 0.95 to obtain the final speech signal.

5. RESULTS AND DISCUSSION

The performance of the designed end-to-end system was measured over real USRP-2900 loopback transmission and evaluated with quantitative metrics. The fundamental performance values obtained are summarized in Table 4.

Table 4. Measured performance metrics.

Metric	Value
Symbol rate	50 kBd (1 bit/symbol)
On-air bit rate	50 kbps
Occupied bandwidth	67.5 kHz (50 kHz × 1.35)
Spectral efficiency	0.74 bps/Hz
Sync correlation SNR	31.5 dB
Measured SNR	17.1 dB
EVM (rms)	14 %
BER (measured)	1.50×10^{-4}
BER (theoretical)	2.23×10^{-24}
Error-free frames	51/52
Encoded bit rate	12.34 kbps
PCM reference rate	128.0 kbps
Compression ratio	10.4 : 1

5.1 Bit Error Rate and Theoretical Agreement

In the actual system, the bit error rate was calculated to be 1.50×10^{-4} (about one erroneous bit among 6656 transmitted bits), while in theory the predicted BER is much smaller, 2.23×10^{-24}, for the measured SNR of 17.1 dB on the theoretical BPSK curve. As expected, there is a discrepancy of about twenty orders of magnitude between the two and this difference does not imply any mistake in the design but instead reveals a key difference between the two cases. For the theoretical BPSK curve, the assumption is made that the channel is a perfect AWGN (additive white Gaussian noise) channel with only the effect of thermal noise. In our measurements, the SNR is calculated based on the error vector magnitude (EVM) of the constellation and hence includes not just thermal noise but also non-Gaussian impairments caused by hardware. These distortions will spread the constellation points in a non-Gaussian way and force some of them over the decision threshold, resulting in more bit errors than what would occur with a pure AWGN channel having the same SNR; therefore, the measured point will lie quite far above and to the right of the ideal curve. The other factor contributing to this is the limited number of transmitted bits (only 6656): a BER of the order of 10^{-4} means that there would be no more than a single bit error, which is insufficient for statistical estimation purposes. The same effect can be noted in variations between successive runs with nominally the same conditions, and obtaining a statistically significant measure of the error rate at this BER would require at least a million of transmitted bits. Understanding this deviation was the most educational part of the work because it showed that an estimated SNR value obtained through the EVM approach is not the same as actual Eb/N0 and that hardware implementations do deviate from theoretical descriptions. Figure 1 illustrates the transmission baseband waveform, the received waveform, original vs reconstructed speech, and spectrogram of the latter. Since there is significant correlation between the original signal (in blue) and the reconstructed signal (in red) on account of the overlapping envelopes, it can be concluded that the model accurately represents the characteristics of the speech. Small variations in the sample-by-sample comparisons should be expected considering the fact that LPC coding works based on the spectral envelope and excitation rather than the sample-by-sample waveform representation. The horizontal high energy bands in the spectrogram represent the formants associated with the speech synthesis, while the vertical gaps represent the silence regions, showing that both spectral envelope and voicing of the speech have been preserved.

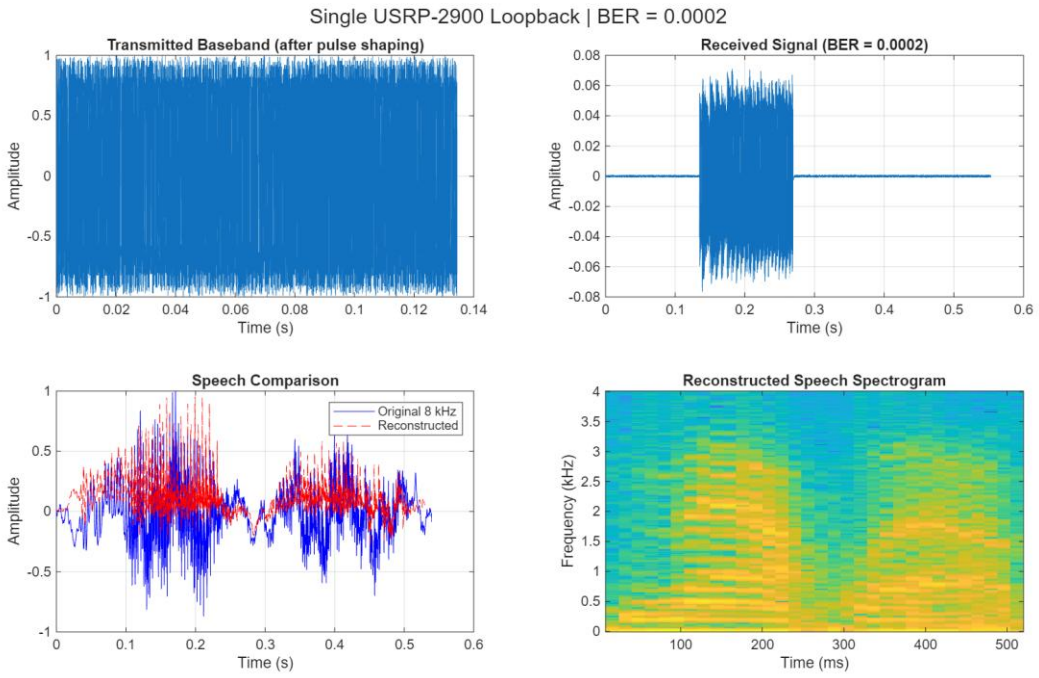


Figure 1. *USRP-2900 loopback diagnostic plots: transmitted baseband signal (top left), received signal (top right), comparison of original/reconstructed speech (bottom left), and the spectrogram of the reconstructed speech (bottom right) (BER = 0.0002).*

5.2 Synchronization Performance and Phase Correction

The SNR of the complex cross-correlation computed for frame synchronization was measured as 31.5 dB. This high value indicates that the preamble peak is very clearly distinguished from the noise floor and that the frame start was detected without error. The strong correlation peak provided both correct timing alignment and reliable resolution of the 180° phase ambiguity. As seen in Figure 2, after phase correction the symbols in the receiver constellation separate clearly into two clusters along the In-phase axis around the ideal ±1 points; this visually confirms the success of the synchronization and phase correction. In the BER–Eb/N0 plot of Figure 2, the measured operating point lies above and to the right of the theoretical BPSK curve, visually reflecting the gap between the EVM-based measured SNR and the ideal AWGN model discussed in Section 5.1. As seen in Figure 2, the power spectrum plots for TX and RX demonstrate that both the spectrums are located in the same frequency band, thereby proving that there is no spectral distortion caused to the signal due to channel impairment; the RX power spectrum is about 30 dB lower than the TX power spectrum, just like before due to path loss.

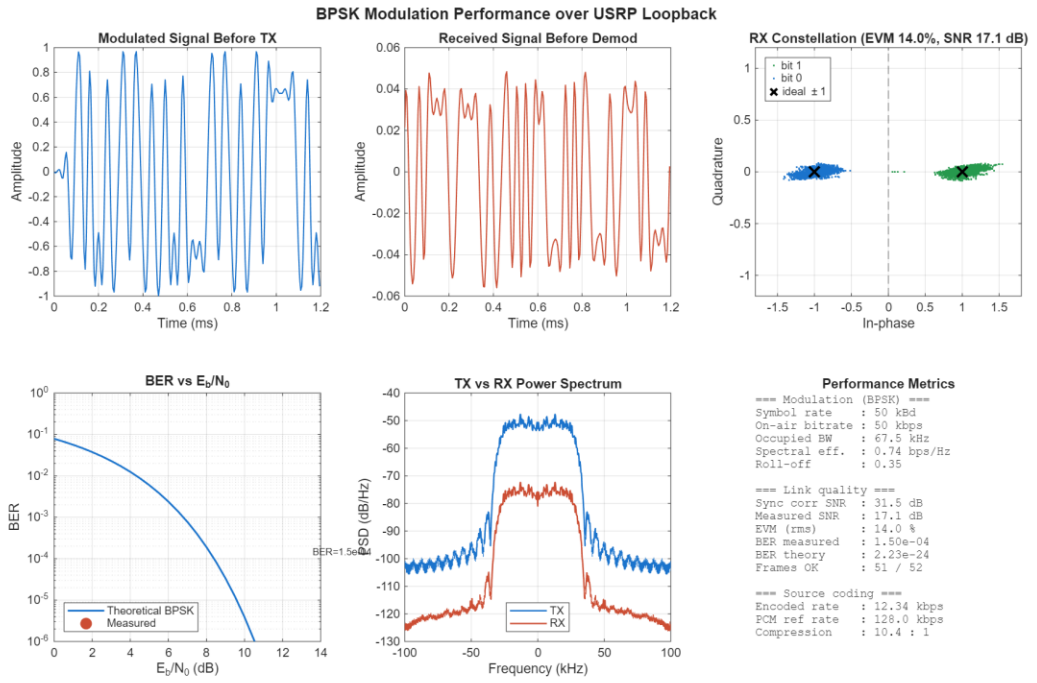


Figure 2. *BPSK performance plots over USRP loopback: TX/RX time signals, receiver constellation (EVM 14%, SNR 17.1 dB), measured-versus-theoretical BER comparison, TX/RX power spectrum, and the performance metrics panel.*

5.3 EVM and Decision-Region Analysis

The measured error vector magnitude (EVM) was obtained as 14.0%. This relatively low EVM value is consistent with the measured SNR of 17.1 dB and indicates that the received symbols cluster tightly around the ideal points. The fact that the bit error rate remained as low as the order of 10^{-4} is a direct consequence of the one-dimensional decision-region structure of BPSK. In BPSK, the decision is made solely by examining the sign of the In-phase axis, so noise scatter along the Quadrature axis does not affect the decision outcome; the bit is decoded correctly as long as the symbols do not cross the decision boundary (the I = 0 axis). Therefore, the two clusters remain clearly separable along the decision axis, producing a very low error rate. Figure 3 shows the modulated bit stream, the BPSK TX signal for the first symbols, and the power spectral density of the transmitted signal; the sharp roll-off of the spectrum at around 33–35 kHz confirms the band-limiting effect of the RRC pulse shaping.

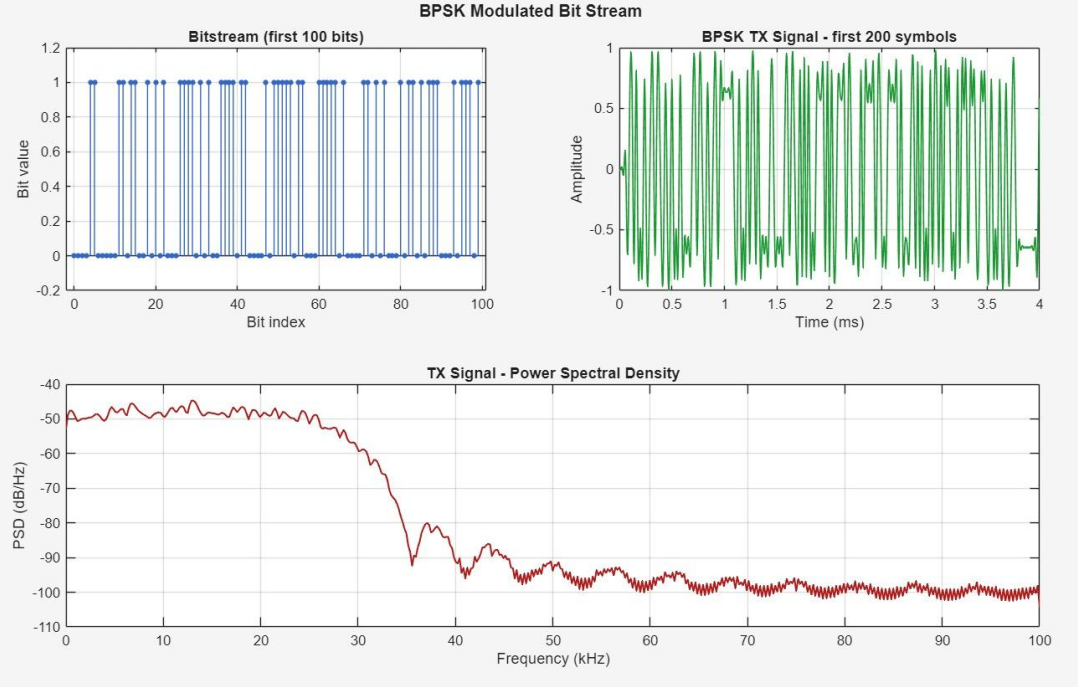


Figure 3. *BPSK-modulated bit stream: the first 100-bit data sequence (top left), the BPSK TX signal for the first 200 symbols (top right), and the power spectral density of the transmitted signal (bottom).*

5.4 Frame Structure and Parameter Distribution

Figure 4 visualizes the frame data structure. Each of the 52 frames carries 128 bits (a total of 52 × 128 = 6656 bits), and the largest portion of the bit budget (112 bits) is allocated to the LSF coefficients representing the spectral envelope. The LSF coefficient map exhibits a smooth variation from frame to frame, reflecting the slowly time-varying nature of the spectral envelope and hence the suitability of the LPC model. Examining the pitch and V/UV plot, the pitch period follows a consistent trend in voiced frames, while in unvoiced frames the scattered values are correctly marked by the V/UV flag. The frame gain curve shows two peaks of energy, with a significant trough in between the two, which precisely relates to the two speech samples, and the gap in between, and follows the energy envelope shown in the speech comparison curve.

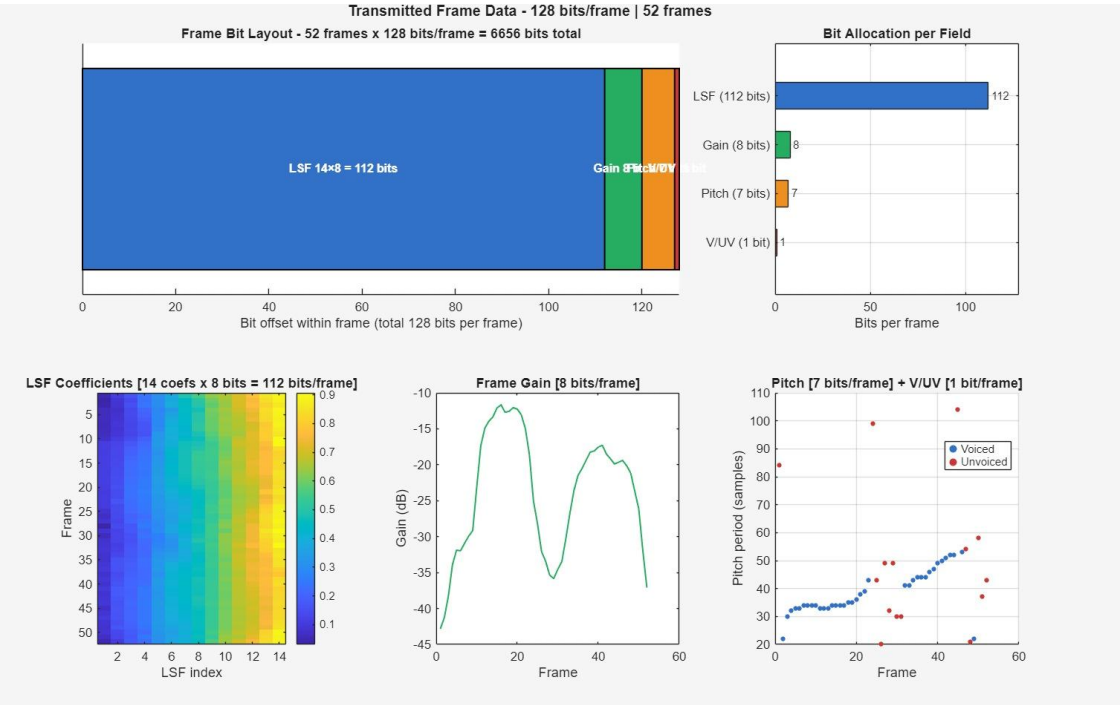


Figure 4. *Transmitted frame data: frame bit layout (128 bits/frame × 52 frames), per-field bit allocation, LSF coefficient map, frame gain (dB), and pitch period + V/UV distribution.*

5.5 Compression Performance and Spectral Efficiency

On the source-coding side, the 128 kbps bit rate that a 16-bit PCM representation would require at 8 kHz was reduced to 12.34 kbps thanks to the parametric LSF representation. This corresponds to a compression ratio of 10.4 : 1. As a direct consequence of the compression, the reduction in the number of bits that must be transmitted positively affected spectral efficiency; the spectral efficiency of the system was measured as 0.74 bps/Hz. The use of LSFs instead of LPC coefficients increased robustness to quantization errors and made it easier to preserve the stability of the synthesis filter. This result quantitatively demonstrates the fundamental benefit of parametric coding over bandwidth-limited channels: the same speech content was transmitted at a bit rate an order of magnitude lower than raw PCM transmission, and therefore using fewer channel resources.

5.6 Future Improvements and Trade-offs

There exist several possibilities for improving the performance of the system. Firstly, the use of channel coding techniques (FEC) would allow obtaining much lower BER values. For instance, an application of LDPC or convolutional coding scheme would help to obtain much lower error rates in case of low SNR levels, although it would result in increasing the level of redundancy and the bit rate correspondingly, and would help to recover the single frame (out of 52) that was distorted due to the channel. This is exactly how the trade-off between efficiency and reliability works for channel coding techniques: increased reliability at the expense of spectral efficiency is gained. Secondly, switching from BPSK to QPSK would result in doubling the bit rate and the spectral efficiency correspondingly while operating within the same bandwith, although QPSK

requires a greater Eb/N0 value for the same BER and is more sensitive to phase distortions comparing to BPSK. Besides, reporting the perceptual quality of speech using objective metrics like PESQ/MOS, and performing predictive quantization on pitch and LSF parameters are some other possible enhancements. Taking into account all these opportunities, an appropriate solution may be found.

6. CONCLUSIONS

In this study, a whole digital communication system, including the processing of the degraded speech signal, parametric encoding, BPSK modulation, and real wireless data transmission and reception at 915 MHz using the USRP-2900 radio, have been developed. The system covers all steps of the algorithm described above, that is, the signal preprocessing, parametric speech encoding (LPC/LSF analysis), quantization, BPSK modulation, real transmission and reception with loopback using USRP hardware, synchronization, demodulation, and LPC synthesis.

Based on the experimental research results, it can be claimed that the developed system performs correctly and meets the requirements set out in the project specifications. First, a low value of measured BER, equal to 1.50×10^{-4} (51 of 52 frames are free from errors), has been attained; the discrepancy between this result and theoretical predictions for BPSK modulation can be explained by the non-ideal SNR value estimation based on the EVM measurement results. Second, the value of the correlation coefficient reached 31.5 dB, which means high performance in frame synchronization and ambiguity elimination. Third, concerning the issues of source coding, it should be pointed out that the achieved compression ratio is 10.4:1 and relatively high spectral efficiency of 0.74 bps/Hz. In such a way, the applicability of parametric encoding is proven in practice. Fourth, in connection with the robustness of the system, it can be mentioned that 51 out of 52 frames were transmitted without any errors and the reconstructed speech had a good quality. Concerning the trade-off between performance and efficiency, it should be stated that the selection of BPSK results in ensuring minimum bit error rate, whereas the relatively low spectral efficiency of 0.74 bps/Hz takes place. Finally, as regards the practical application of the considered techniques, it can be concluded that the usage of both communication and DSP methods will enable the creation of the end-to-end system which guarantees intelligible reconstructed speech and can be measured in terms of performance. Such an end-to-end system represents a modifiable platform that can be improved, for example, using FEC and QPSK.

7.REFERENCES

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